

STAR Pattern

1. Square

```
C++ square.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the square matrix: ";
7      cin >> n;
8
9      for(int i = 0; i < n; i++){
10         for(int j = 0; j < n; j++){
11             cout << "*" << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ square.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the square matrix: 5
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

PS C:\Users\manoj\OneDrive\Documents\DSA C++> █

2. Hollow Square

```
C++ HS.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the hollow square matrix: ";
7      cin >> n;
8
9      for(int i = 0; i < n; i++){
10         for(int j = 0; j < n; j++){
11             if(i == 0 || i == n-1 || j == 0 || j == n-1){
12                 cout << "*" << " ";
13             }
14             else{
15                 cout << "  " << " ";
16             }
17         }
18         cout << endl;
19     }
20     return 0;
21 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ HS.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the hollow square matrix: 5
* * * * *
*      *
*      *
*      *
* * * * *
```

3. Triangle

```
C++ tri.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the triangle: ";
7      cin >> n;
8
9      for(int i = 0; i < n; i++){
10         for(int j = 0; j <= i; j++){
11             cout << "*" << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ tri.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the triangle: 5
*
* *
* * *
* * * *
* * * * *
```

4. Inverted Triangle

```
C++ intri.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the Iverted triangle: ";
7      cin >> n;
8
9      for(int i = n - 1; i >= 0; i--){
10         for(int j = 0; j <= i; j++){
11             cout << "*" << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ intri.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the Iverted triangle: 5
* * * * *
* * * *
* * *
* *
*
```

5. Pyramid

```
C++ pyr.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the pyramid: ";
7      cin >> n;
8      //left half pyramid
9      for(int i = 0; i < n; i++){
10         for(int j = 1; j <= n - i - 1; j++){
11             cout << " ";
12         }
13
14         for(int j = 1; j <= i + 1; j++){
15             cout << "* ";
16         }
17
18
19
20         //right half pyramid
21
22         for(int j = 1; j <= i; j++){
23             cout << "* ";
24         }
25         cout << endl;
26     }
27     return 0;
28 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ pyr.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the pyramid: 5
    *
  * * *
 * * * * *
* * * * * *
* * * * * * *
```

6. Diamond

```
C++ diam.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the Diamond: ";
7      cin >> n;
8      //upper half diamond
9      for(int i = 0; i < n; i++){
10         for(int j = 1; j <= n - i-1; j++){
11             cout << " ";
12         }
13
14         for(int j = 1; j <= i + 1; j++){
15             cout << "* ";
16         }
17
18
19         for(int j = 1; j <= i; j++){
20             cout << "* ";
21         }
22         cout << endl;
23     }
24
25     //lower half diamond
26     for(int i = n - 2; i >= 0; i--){
27         for(int j = 1; j <= n - i-1; j++){
28             cout << " ";
29         }
30
31         for(int j = 1; j <= i + 1; j++){
32             cout << "* ";
33         }
34
35         for(int j = 1; j <= i; j++){
36             cout << "* ";
37         }
38         cout << endl;
39     }
40     return 0;
41 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ diam.cpp
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the pyramid: 5
      *
    * * *
  * * * * *
* * * * * *
* * * * * *
  * * * * *
    * * *
      *
```

7. Half Diamonds

```

C++ hldiam.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the half Diamond: ";
7      cin >> n;
8      //upper half diamond
9      for(int i = 0; i < n; i++){
10         for(int j = 0; j <= i; j++){
11             cout << "* ";
12         }
13         cout << endl;
14     }
15     //lower half diamond
16     for(int i = n - 2; i >= 0; i--){
17         for(int j = 0; j <= i; j++){
18             cout << "* ";
19         }
20         cout << endl;
21     }
22     return 0;
23 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ hldiam.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the half Diamond: 5
*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

8. Half Diamnod Inverted

```

C++ indiam.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the Inverted Diamond: ";
7      cin >> n;
8
9      //lower half diamond
10     for(int i = n - 1; i >= 0; i--){
11         for(int j = 1; j <= n - i; j++){
12             cout << " ";
13         }
14
15         for(int j = 1; j <= i + 1; j++){
16             cout << "* ";
17         }
18
19         for(int j = 1; j <= i; j++){
20             cout << "* ";
21         }
22         cout << endl;
23     }
24     return 0;
25 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ indiam.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the Inverted Diamond: 5
*
* *
* *
* *
* *

```

NUMBER Patterns

1. Square

```
C++ num1.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the square matrix: ";
7      cin >> n;
8
9      for(int i = 0; i < n; i++){
10         for(int j = 0; j < n; j++){
11             cout << "1 ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num1.cpp
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the square matrix: 5
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

2. Incrementing

```
C++ num2.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the square matrix: ";
7      cin >> n;
8
9      for(int i = 1; i <=n; i++){
10         for(int j = 1; j <= n; j++){
11             cout << j << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num2.cpp
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the square matrix: 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

3. Right Triangle

```
num3.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the triangle: ";
7      cin >> n;
8
9      for(int i = 1; i <= n; i++){
10         for(int j = 1; j <= i; j++){
11             cout << j << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num3.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the triangle: 5
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

4. Inverted Diamond

```
num4.cpp > main()
1  #include<iostream>
2  using namespace std;
3
4  int main(){
5      int n;
6      cout << "Enter the size of the Inverted Diamond: ";
7      cin >> n;
8
9      for(int i = 3; i <= 7; i++){
10         for(int j = 7; j >= i; j--){
11             cout << i << " ";
12         }
13         cout << endl;
14     }
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num4.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the Inverted Diamond: 5
3 3 3 3 3
4 4 4 4
5 5 5
6 6
7
```

5. Sandwich Pattern

```
C++ num5.cpp > @main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cout << "Enter the size of the Sandwich Pattern: ";
7      cin >> n;
8
9      // Upper part
10     for (int i = 0; i < 2; i++) {
11         // Spaces
12         for (int j = 0; j < i; j++)
13             cout << " ";
14
15         // Numbers
16         for (int j = 0; j < 5 - 2 * i; j++)
17             cout << n + i << " ";
18
19         cout << endl;
20     }
21
22     // Middle row
23     for (int i = 0; i < 2; i++)
24         cout << " ";
25     cout << 5 << endl;
26
27     // Lower part
28     for (int i = 1; i >= 0; i--) {
29         // Spaces
30         for (int j = 0; j < i; j++)
31             cout << " ";
32
33         // Numbers
34         for (int j = 0; j < 5 - 2 * i; j++)
35             cout << n + i << " ";
36
37         cout << endl;
38     }
39
40     return 0;
41 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num5.cpp
PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the Sandwich Pattern: 3
3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3
```

6. Incrementing Diamond Pattern


```
C++ num6.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int start = 3;
6      int n ;
7      cout << "Enter the size of the Incrementing Diamond Pattern: ";
8      cin >> n;
9
10     // Upper half
11     for (int i = 1; i <= n; i++) {
12         int num = start + i - 1;
13
14         for (int j = 1; j <= i; j++) {
15             cout << num << " ";
16         }
17         cout << endl;
18     }
19
20     // Lower half
21     for (int i = n - 1; i >= 1; i--) {
22         int num = start + i - 1;
23
24         for (int j = 1; j <= i; j++) {
25             cout << num << " ";
26         }
27         cout << endl;
28     }
29
30     return 0;
}

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> g++ num6.cpp
● PS C:\Users\manoj\OneDrive\Documents\DSA C++> .\a.exe
Enter the size of the Incrementing Diamond Pattern: 5
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```